TreY/TreZ Trehalose Biosynthesis from α-Glucan in *Pseudomonas putida* KT2440

Commissioned Module/Pathway/Taxon Review Brief

- **Target taxon:** *Pseudomonas putida* KT2440 (organism code PSEPK; NCBI taxon 160488; proteome UP000000556)
- **Module under review:** TreY/TreZ trehalose biosynthesis from α-1,4-glucan (KEGG map ppu00500, *Starch and sucrose metabolism*, local bucket `kegg:ppu00500`)
- **Module area:** other_kegg_pathway
- **Verdict at a glance:** Module **SATISFIABLE** / **COVERED**, with TreY flagged **candidate_uncertain** and promoted to full gene review.

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1. Executive Summary

The two-reaction **TreY/TreZ** module — in which TreY (maltooligosyltrehalose synthase, EC 5.4.99.15) isomerizes the reducing-end α-1,4 linkage of an α-glucan into an α,α-1,1 (trehalose) linkage, and TreZ (malto-oligosyltrehalose trehalohydrolase, EC 3.2.1.141) hydrolyzes the resulting maltooligosyltrehalose to release free trehalose — is **gene-complete and satisfiable** in *Pseudomonas putida* KT2440. Both catalytic steps are encoded within a single contiguous glycogen/GlgE/trehalose locus on the chromosome: **treZ = PP_4051 (Q88FN8)** and **treY = PP_4053 (Q88FN6)**, flanking `glgA (PP_4050)`, `malQ (PP_4052)`, and physically clustered with `glgX (PP_4055)`, `glgB (PP_4058)`, `treSB/Mak (PP_4059)`, and `glgE (PP_4060)`. The polymeric α-glucan substrate on which TreY/TreZ act is supplied *in cis* by the glycogen/GlgE machinery of the same locus. This co-localization is the strongest single line of evidence for module satisfiability.

The module's biological importance in KT2440 is **elevated rather than redundant**, because the canonical OtsA/OtsB (trehalose-6-phosphate synthase/phosphatase) route is **entirely absent** from the reference proteome. We confirmed this absence twice and by two independent methods: (i) name/EC-based screens returned zero hits for OtsA (EC 2.4.1.15), OtsB (EC 3.1.3.12), any `ots*` gene, or trehalase (EC 3.2.1.28); and (ii) a name-independent InterPro domain screen returned zero proteins carrying the OtsA glycosyltransferase-20 family domain

(IPR001830/IPR012766) or the trehalose-6-phosphate phosphatase domain (IPR003337). Because the domain search does not depend on gene naming, a mis-named or divergent OtsA/OtsB is effectively excluded. TreYZ, together with a genuine standalone TreS (treSA/PP_2918) and the GlgE-pathway maltokinase-TreS fusion (treSB/PP_4059), constitute the trehalose-related enzymology of this strain.

For curation, the actionable conclusions are: **mark both module steps covered**, but flag **step 1 (TreY) as candidate_uncertain** and promote it to full `fetch-gene` review because UniProt records it at protein-existence level PE=4 ("Predicted") with **no EC number and no catalytic-activity reaction populated**, in contrast to TreZ (PE=3, EC 3.2.1.141, populated reaction). The generic module boundary — which keeps α-glucan synthesis, GlgE remodeling, the TreS interconversion, the OtsAB route, and trehalose degradation outside the module — is **correct for this organism** and needs no revision. The OtsAB branch of any parent trehalose-biosynthesis supermodule should be marked **not_expected_in_target_taxon** for KT2440.

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2. Target-Organism Pathway Definition

2.1 Exact process included in the module

The module comprises exactly two enzymatic reactions acting on a preformed α-1,4-glucan (glycogen-like polymer or maltooligosaccharide):

1. **Maltooligosyltrehalose formation** — TreY (EC 5.4.99.15; (1,4)-α-D-glucan 1-α-D-glucosylmutase) intramolecularly transglycosylates the terminal α-1,4 linkage at the reducing end into an α,α-1,1 linkage, producing **maltooligosyltrehalose** (a trehalose cap on the end of the glucan chain).
2. **Trehalose release** — TreZ (EC 3.2.1.141; 4-α-D-((1→4)-α-D-glucano)trehalose trehalohydrolase) hydrolyzes the α-1,4 glucosidic bond adjacent to the newly formed trehalose unit, releasing **free trehalose** and a shortened α-glucan that can re-enter the cycle.

2.2 Neighboring processes to keep separate

The following are **outside** the module boundary and should not be conflated with TreYZ steps:

- **α-Glucan synthesis and remodeling** — glycogen synthase `glgA` (PP_4050, EC 2.4.1.21), branching enzyme `glgB` (PP_4058, EC 2.4.1.18), debranching `glgX` (PP_4055, EC 3.2.1.33), amylomaltase `malQ` (PP_4052, EC 2.4.1.25), glucan phosphorylase `glgP` (PP_5041, EC 2.4.1.1). These *supply and recycle* the substrate but are not trehalose-forming steps.

- **The GlgE pathway** — maltokinase/TreS fusion `treSB` (PP_4059, EC 2.7.1.175 / 5.4.99.16) and maltosyltransferase `glgE` (PP_4060, EC 2.4.99.16), which build α-glucan from α-maltose-1-phosphate.
- **Alternative trehalose routes** — the OtsA/OtsB (trehalose-6-phosphate) route and the standalone TreS (maltose⇌trehalose isomerase) route.
- **Trehalose degradation** — trehalase (EC 3.2.1.28).
- **Peripheral central-carbon and cell-envelope genes** — `glk` (glucokinase), `bglX` (β-glucosidase), `pgi1` / `pgi2` (phosphoglucose isomerase), `bcsA` (cellulose synthase, β-glucan), `pgm` / `cpsG` / `algC` (phosphogluco/mannomutases), `galU` (UDP-glucose pyrophosphorylase). These are members of KEGG map ppu00500/adjacent maps but are **not** TreYZ steps.

2.3 Alternate names and database definitions

- KEGG pathway **ppu00500** = "Starch and sucrose metabolism" (the parent map; TreY/TreZ are two nodes within it).
- The TreY/TreZ route is also referred to as the **"OtsAB-independent" or "maltooligosyltrehalose (MOT) pathway"** of trehalose biosynthesis, distinct from the OtsAB and TreS pathways.
- Enzyme synonyms: TreY = "malto-oligosyltrehalose synthase / MTSase"; TreZ = "malto-oligosyltrehalose trehalohydrolase / MTHase."
- InterPro provides **dedicated family signatures**: IPR012767 (Trehalose_TreY) and IPR012768 (Trehalose_TreZ), both built on the GH13 α-amylase superfamily (IPR006047).

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3. Expected Step Model

Step	Enzyme (activity)	Expected EC	KT2440 gene	Status
1. Maltooligosyltrehalose formation	TreY — (1,4)-α-D-glucan 1-α-D-glucosylmutase	5.4.99.15	**treY** / **PP_4053** / **Q88FN6**	**Covered but candidate_uncertain**
2. Trehalose release	TreZ — 4-α-D-(1→4)-α-D-glucanotrehalose trehalohydrolase	3.2.1.141	**treZ** / **PP_4051** / **Q88FN8**	**Covered**
(Substrate supply, *outside module*)	α-glucan synthesis/ remodeling	2.4.1.21 / 2.4.1.18 / 2.4.99.16	glgA, glgB, glgE, malQ	Present (context)

Both required steps map cleanly onto KT2440 genes. There are **no missing steps** in the two-reaction core; the only uncertainty is the depth of annotation evidence for TreY (see §4–§5).

```
α-1,4-glucan (glycogen-like, supplied by glgA/glgB/glgE/malQ)
           │
           │  TreY  (PP_4053, EC 5.4.99.15)   ── STEP 1
           ▼         reducing-end α-1,4 → α,α-1,1 isomerization
maltooligosyltrehalose  (trehalose-capped glucan)
           │
           │  TreZ  (PP_4051, EC 3.2.1.141)   ── STEP 2
           ▼         hydrolysis adjacent to trehalose cap
free trehalose  +  shortened α-glucan ──┐ (recycles)
                                        └──► back to substrate pool
```

4. Candidate Genes and Evidence

4.1 Core module genes (high priority)

treZ — PP_4051 (Q88FN8) — Malto-oligosyltrehalose trehalohydrolase (Step 2). This is the **best-supported** gene in the module. UniProt records it at protein-existence level **PE=3 "Inferred from homology"**, with **EC 3.2.1.141** and a **fully populated catalytic-activity reaction** (hydrolysis of the (1→4)-α-D-glucosidic linkage in maltooligosyltrehalose to yield trehalose + α-glucan). It carries the dedicated InterPro **TreZ family signature IPR012768** (Trehalose_TreZ). Evidence type: homology + dedicated family assignment + populated EC/reaction. Curation caveat: no experimental (PE=1) evidence *in the target strain*, but the annotation is internally consistent and family-specific. **Mark covered.**

treY — PP_4053 (Q88FN6) — Maltooligosyl trehalose synthase (Step 1). The 924-aa protein carries the dedicated InterPro **TreY family signature IPR012767** and the GH13 catalytic domain (IPR006047) — architecturally a bona fide TreY. However, UniProt records it at **PE=4 "Predicted"** with **no EC number and no catalytic-activity reaction populated**, making its evidence base weaker than TreZ's. The protein length (924 aa) is consistent with a large GH13 maltooligosyltrehalose synthase. Evidence type: family-signature homology only. Curation caveat: the family assignment is strong, but the missing EC/reaction and PE=4 status mean the functional call has not been formally curated. **Mark covered but candidate_uncertain; promote to full review.**

4.2 Substrate-supply / GlgE-pathway context genes (present, outside module)

Gene	Locus	UniProt	Annotation	EC	Role relative to module
glgA	PP_4050	Q88FN9	Glycogen synthase	2.4.1.21	Supplies α-glucan substrate
malQ	PP_4052	Q88FN7	4-α-glucanotransferase (amylomaltase)	2.4.1.25	Glucan remodeling
glgX	PP_4055	Q88FN4	Glycogen debranching enzyme	3.2.1.33	Glucan remodeling
glgB	PP_4058	Q88FN1	1,4-α-glucan branching enzyme	2.4.1.18	Glucan branching
treSB	PP_4059	Q88FN0	Maltokinase + TreS (bifunctional)	2.7.1.175 / 5.4.99.16	GlgE-pathway maltokinase
glgE	PP_4060	Q88FM9	Maltose-1-P maltosyltransferase	2.4.99.16	α-glucan synthesis
glgP	PP_5041	Q88CY8	α-1,4 glucan phosphorylase	2.4.1.1	Glucan degradation

The GlgE-pathway gene set is complete and co-located with treY/treZ, defining the pathway neighborhood. In *M. tuberculosis* and *S. venezuelae*, the TreS-Pep2-GlgE and GlgC-GlgA routes converge on **α-maltose-1-phosphate** to build branched α-glucan — the polymeric substrate on which TreY/TreZ act (PMID: 27513637; PMID: 27121970).

4.3 TreS-family paralogs (resolved)

A key curation subtlety is the presence of **two distinct TreS-family paralogs** that must not be conflated:

- **treSA — PP_2918 (Q88IT1), 688 aa, PE=4** — carries the TreS family signature **IPR012665 (Trehalose_synth)** + GH domain (IPR017853) and **no maltokinase domain**. This is a **classic standalone trehalose synthase** (maltose ⇌ trehalose isomerase). A TreS from *P. putida* ATCC47054 has been used industrially to convert maltose to trehalose in one step (PMID: 29602391), consistent with a functional standalone TreS existing in the species.

- **treSB — PP_4059 (Q88FN0), 1106 aa, PE=3** — carries the TreS/α-amylase N-domain (IPR012810) **AND** the dedicated TreS-maltokinase C-domain (**IPR012811**, TreS_maltokin_C_dom) plus Mak_N_cap (IPR040999) and a kinase-like domain (IPR011009), with **both reactions populated** (D-maltose + ATP → α-maltose-1-phosphate; D-maltose → trehalose). A proteome scan for IPR012810 returned only treSB. This is a **genuine bifunctional TreS-maltokinase (Pep2/Mak-type)** whose in-operon role is to supply the GlgE pathway with maltose-1-phosphate — **not** a canonical stand-alone trehalose synthase.

4.4 Peripheral central-carbon genes (not module steps)

`glk` (PP_1011), `bglX` (PP_1403), `cpsG` (PP_1777), `pgi1` (PP_1808), `bcsA` (PP_2635, cellulose synthase → β-glucan, not α-glucan), `pgm` (PP_3578), `galU` (PP_3821), `pgi2` (PP_4701), and `algC` (PP_5288) are members of ppu00500 or adjacent maps but are **peripheral to** the TreYZ module. They participate in general hexose-phosphate interconversion, UDP-glucose supply, or cell-envelope polysaccharide synthesis, and should be excluded from module satisfiability accounting.

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5. Gaps, Ambiguities, and Likely Over-Annotations

5.1 The single genuine gap: TreY annotation depth

The only substantive gap is the **weak annotation evidence for TreY (PP_4053)**. Despite carrying the dedicated TreY InterPro family signature, it is PE=4 ("Predicted") with no EC number and no catalytic-activity reaction populated in UniProt. This is an **annotation-completeness gap, not a gene-presence gap** — the gene is clearly present and family-assigned, but its functional call has not been formally curated to the level of TreZ. This is why it is flagged candidate_uncertain and promoted to full review.

5.2 Absence of the OtsAB route (confirmed, not a gap in the module)

The OtsA/OtsB trehalose-6-phosphate route is **truly absent** from KT2440, confirmed by two independent methods:

1. **Name/EC screen** of proteome UP000000556 — zero hits for OtsA (EC 2.4.1.15), OtsB (EC 3.1.3.12), any `ots*` gene, and trehalase (EC 3.2.1.28).

2. **Domain-level screen** — zero proteins with the OtsA glycosyltransferase-20 family domain (IPR001830 / IPR012766) or the trehalose-6-phosphate phosphatase domain (IPR003337).

Because the second method is name-independent, a mis-annotated or divergently named OtsA/OtsB is effectively excluded. This is a **biologically real absence**, and it raises the physiological importance of the TreYZ route: KT2440 accumulates trehalose as a desiccation protectant (PMID: 31323038), so an active OtsAB-independent trehalose source is required.

5.3 Potential over-annotation / paralog ambiguity to watch

- **treSB (PP_4059)** could be over-simplified in downstream databases as a plain "trehalose synthase." Its dedicated maltokinase C-domain (IPR012811) shows it is really a **bifunctional GlgE-pathway maltokinase**; annotation pipelines that collapse it to "TreS" would misrepresent its in-operon role.
- **treSA vs treSB** must be kept distinct — both hit TreS-family signatures but have different architectures and roles.
- **Broad EC mappings**: several substrate-supply genes carry broad GH13 or transferase EC numbers that overlap the trehalose enzymes' superfamily; care is needed not to let superfamily-level EC promiscuity leak into TreYZ step assignments.

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6. Module and GO-Curation Recommendations

Module step	Recommended status	Rationale
Step 1 — Maltooligosyltrehalose formation (TreY)	**covered / candidate_uncertain**	Present (PP_4053, dedicated TreY family IPR012767) but PE=4, no EC/reaction populated → promote to full review
Step 2 — Trehalose release (TreZ)	**covered**	PP_4051, PE=3, EC 3.2.1.141, populated reaction, dedicated TreZ family IPR012768
OtsAB branch (parent supermodule)	**not_expected_in_target_taxon**	Absent by both name/EC and domain-level screens

Module boundary assessment: The generic module boundary is **correct for KT2440** and needs no revision. Keeping α-glucan synthesis, GlgE remodeling, TreS interconversion, OtsAB, and trehalose degradation outside the module accurately reflects the biology, where these are distinct (if physically co-located) processes.

GO / new-document needs: No new module document or GO term request is strictly required. The two required GO/EC activities already exist (EC 5.4.99.15 for TreY; EC 3.2.1.141 for TreZ). The main curation action is to **populate the EC number (5.4.99.15) and catalytic-activity reaction for TreY (Q88FN6)** upon full review, upgrading it from PE=4 to a curated homology call, so that automated module satisfiability tools register step 1 as fully covered.

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7. Genes to Promote to Full Review

1. **treY / PP_4053 / Q88FN6 (HIGH PRIORITY).** Promote to full `fetch-gene` review to confirm the maltooligosyltrehalose-synthase function, assign EC 5.4.99.15, and populate the catalytic reaction. Rationale: it is the one module step whose evidence (PE=4, no EC/reaction) lags behind its family assignment.
2. **treZ / PP_4051 / Q88FN8 (optional confirmatory).** Already well-annotated (PE=3, EC 3.2.1.141, populated reaction); a light review to confirm the family call would fully close the module.
3. **treSB / PP_4059 / Q88FN0 (context, medium).** Worth a review to ensure downstream databases retain its bifunctional TreS-maltokinase (GlgE-pathway) identity and do not over-collapse it to "trehalose synthase."

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8. Mechanistic Model / Interpretation

The picture that emerges is of a **single, tightly organized chromosomal locus** in *P. putida* KT2440 that couples glycogen/α-glucan metabolism directly to trehalose production:

```
Chromosomal locus (P. putida KT2440):

 PP_4050   PP_4051   PP_4052   PP_4053   ... PP_4055  PP_4058  PP_4059   PP_4060
  glgA       treZ      malQ      treY          glgX     glgB     treSB     glgE
 (synth)  (STEP 2)  (remodel) (STEP 1)       (debr.) (branch) (Mak-TreS) (GlgE)
   │         ▲          │         │             │        │        │         │
   └─────────┼──────────┴─────────┼─────────────┴────────┴────────┴─────────┘
             │   α-GLUCAN SUBSTRATE POOL (supplied/recycled in cis)
             │                     │
  trehalose ◄── TreZ ◄── maltooligosyltrehalose ◄── TreY ◄── α-1,4-glucan
```

The substrate for TreY — a branched, glycogen-like α-1,4-glucan — is manufactured and maintained *in cis* by the same locus (glgA/glgB synthesis and branching; glgE/treSB building from maltose-1-phosphate; malQ/glgX/glgP remodeling and turnover). TreY then caps the reducing end with a trehalose unit, and TreZ liberates free trehalose. Because KT2440 has **no OtsAB route**, this MOT (maltooligosyltrehalose) pathway is the principal *de novo* trehalose-synthesis route from glucan, complemented by the standalone TreS (treSA) that can interconvert maltose and trehalose. This architecture is physiologically coherent: trehalose is a validated desiccation/osmotic protectant in KT2440 (PMID: 31323038), and coupling its synthesis to the glycogen store provides an on-demand protectant reservoir.

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9. Evidence Base

PMID	Title (short)	How it supports the findings
27513637	*Metabolic Network for Biosynthesis of α-Glucans Required for Virulence of M. tuberculosis*	"There is an unexpected convergence of the TreS-Pep2 and GlgC-GlgA pathways that both generate α-maltose-1-phosphate." Documents that both routes feed α-maltose-1-phosphate into GlgE-dependent α-glucan synthesis — the substrate pool supplying TreY/TreZ — supporting the module boundary between α-glucan supply and the TreYZ steps.
27121970	*Developmental delay in a Streptomyces venezuelae glgE null mutant …*	"The GlgE pathway is thought to be responsible for the conversion of trehalose into a glycogen-like α-glucan polymer in bacteria." Confirms the GlgE-pathway genes co-located with treY/treZ interconvert trehalose and α-glucan, defining the neighboring process bordering the TreYZ module.
31323038	*Desiccation-induced viable but nonculturable state in P. putida KT2440*	"The BSR in the presence of nonreducing disaccharides, such as trehalose, was high after 15 days of desiccation stress." Direct KT2440 evidence that trehalose is a physiologically relevant protectant, supporting the biological importance of an active trehalose-biosynthesis route (TreYZ) in this strain.
29602391	*A process for production of trehalose by recombinant trehalose synthase …*	A TreS from *P. putida* ATCC47054 converts maltose to trehalose in one step, consistent with a functional standalone TreS (treSA) existing in the species and complementing the TreYZ route.

Supporting molecular evidence is drawn from UniProt (proteome UP000000556) and InterPro family/domain assignments: dedicated TreY (IPR012767) and TreZ (IPR012768) family signatures; the GlgE domain (IPR026585); the TreS-maltokinase C-domain (IPR012811); and the absence of OtsA (IPR001830/IPR012766) and OtsB (IPR003337) domains.

10. Limitations and Knowledge Gaps

- **No experimental (PE=1) evidence in the target strain** for either TreY or TreZ. Both functional calls are homology/family-based. Direct enzymatic assays or knockout phenotypes in KT2440 have not been reported in the literature reviewed here.
- **TreY annotation incompleteness.** TreY is PE=4 with no EC/reaction populated; while its family signature is strong, formal functional confirmation is pending.
- **Regulation and flux not addressed.** This review establishes gene presence and module satisfiability; it does not quantify trehalose flux through TreYZ vs TreS under specific conditions (desiccation, osmotic stress, carbon source).
- **Literature is sparse for KT2440-specific trehalose enzymology.** Much mechanistic transfer relies on *M. tuberculosis* and *S. venezuelae* (GlgE pathway) — strong at the pathway-architecture level but not a substitute for direct KT2440 biochemistry.
- **Substrate specificity of treSB** as a maltokinase vs TreS in vivo is inferred from domain architecture, not measured in KT2440.

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11. Proposed Follow-up Experiments / Actions

Curation actions (immediate): 1. Promote **treY (PP_4053/Q88FN6)** to full `fetch-gene` review; assign EC 5.4.99.15 and populate the catalytic-activity reaction based on the IPR012767 family call. Upgrade module step 1 from candidate_uncertain to covered. 2. Confirm **treZ (PP_4051/Q88FN8)** family call (optional, light review). 3. Annotate **treSB (PP_4059)** explicitly as a bifunctional TreS-maltokinase (GlgE pathway) to prevent over-collapse to "trehalose synthase." 4. Mark the OtsAB branch of any parent trehalose-biosynthesis supermodule as **not_expected_in_target_taxon** for KT2440.

Experimental / expert questions (to resolve gaps): 5. **Knockout phenotyping:** Construct ΔtreY, ΔtreZ, and ΔtreY ΔtreSA double mutants in KT2440 and measure intracellular trehalose under desiccation/osmotic stress. If TreYZ is the principal route, ΔtreY or ΔtreZ should show a substantial trehalose-synthesis defect that ΔtreSA alone does not. 6. **Enzyme assays:** Express and assay recombinant TreY (PP_4053) and TreZ (PP_4051) on glycogen/maltooligosaccharide substrates to confirm EC 5.4.99.15 and EC 3.2.1.141 activities directly in the target strain background. 7. **Flux/regulation:** Use ^{13}C tracing or targeted metabolomics under osmotic/desiccation stress to quantify the relative contributions of TreYZ, TreS (treSA), and any residual routes to the trehalose pool. 8. **Confirm OtsAB absence functionally:** Verify that no trehalose-6-phosphate synthase activity is detectable in KT2440 lysates, corroborating the genomic/domain-level absence.

Appendix: Consolidated Findings (from investigation)

- **F001** — Both TreY/TreZ module steps are encoded by a contiguous glgE/glycogen/trehalose locus (PP_4050–PP_4060) in KT2440; treZ carries EC 3.2.1.141 + IPR012768, treY carries IPR012767 + GH13.
- **F002** — TreY (PP_4053) evidence (PE=4, no EC/reaction) is weaker than TreZ (PE=3, EC + reaction) → promote to full review.
- **F003** — Co-located GlgE-pathway gene set defines the module boundary; convergence on α-maltose-1-phosphate supplies the α-glucan substrate (PMID: 27513637, PMID: 27121970).
- **F004** — TreYZ is the principal trehalose-synthesis route; OtsAB absent from the proteome (zero EC/name hits).
- **F005** — Domain-level InterPro scan independently confirms complete absence of OtsAB (IPR001830/IPR012766/IPR003337 = 0).
- **F006** — Two distinct TreS paralogs: treSA/PP_2918 standalone TreS; treSB/PP_4059 bifunctional TreS-maltokinase.
- **F007** — Final verdict: module covered/satisfiable; TreY candidate_uncertain; OtsAB not_expected_in_target_taxon; generic boundary correct.
